

Lecture 1

→ Coefficient matrix notation for n -variable linear equation.

$$\begin{bmatrix} a_0 & a_1 & \dots & a_n \\ b_0 & b_1 & \dots & \\ \vdots & & & \end{bmatrix}_{m \times n} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_m \end{bmatrix}$$

$m \rightarrow$ number of equations
 $n \rightarrow$ number of variables
 [mostly $m=n$]

eg

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y &= 3 \end{aligned}$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

A X b

→ Row picture of this matrix would be looking at each n-var equation

eg $a_0 x_0 + a_1 x_1 + \dots + a_n x_n = c_0$

Basically multiplying the row of coefficient matrix with the corresponding var column.

→ Column picture would be seeing the problem after separating the vars.
eg.

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Basically what is the linear combination of (x, y) that would lead to the correct solution

→ Another idea to be explored in future would be what are the solutions we can reach with

$$x A + y B$$

→ Another way could be what kind of "space" it fills when there are less no. of equations than variables.

For eg.

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 4 \end{bmatrix} + z \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Now for this we won't have fixed values of x, y, z since we only have 2 equations but what kind of space we fill?

→ Column picture can be seen as how do I combine these vectors to get the desired result. Just a different way to see a problem

→ Column picture example for 3 equations 3 variables.

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

→ The question is $AX = b$ going to always give us a solution. ✓?

3 var

- Is the AX going to fill the whole 3d space or 2d plane for 2 var

- Is AX going to fill the whole vector space?

→ The answer to these questions depend on the columns.
"Dependency" in columns would lead to unreachable space.

- Eg All cols in same plane in 3d would never get out of that plane.

- Eg. One column is directly a multiple of another.

→ Basically row picture and column picture are both correct ways to do a matrix multiplication.

$$\bullet \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 4 \cdot \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \cdot 3 & + & 4 \cdot 4 \\ 1 \cdot 3 & + & 5 \cdot 4 \end{bmatrix} \cdot$$

_____ X _____ X _____

Summary

→ Two ways to see a problem of linear equations -

① Row picture : Each equation expanded
"Traditional"

② Column picture : Seeing the problem
as vectors of coefficients
Good for more vars.

→ Also viewed the concept of solution space a linear equation fills and also whether a solution exists.

→ Dependency between cols $\Rightarrow$ no solution
(if #equations $\neq$ # vars)

